

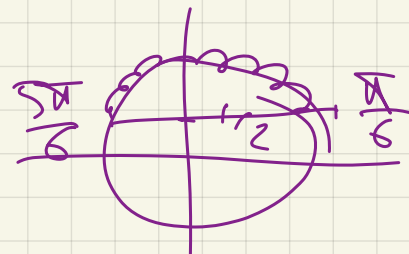


9

$$\sqrt{1+t} = 7 \text{ so } \frac{1+t}{4} > 3.5, \quad \underline{\underline{0 \leq t \leq 2}}$$

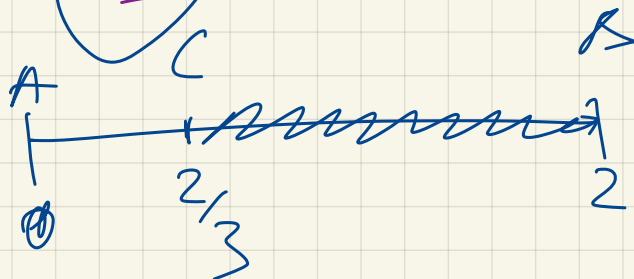
$$\sin \frac{1+t}{4} > \frac{1}{2}$$

$$\frac{5\pi}{6} > \frac{1+t}{4} > \frac{\pi}{6}$$



$$\frac{10}{3} = \frac{20}{6} > t > \frac{4}{6} = \frac{2}{3}$$

$$2 \geq t \geq 0$$



$$AB = 2$$

$$CB = 2 - \frac{2}{3} = \frac{4}{3}$$

$$\frac{CB}{AB} = \frac{\frac{4}{3}}{2} = \frac{2}{3} = 0,67$$

$$\frac{\cos 3x}{2 \sin x + \sqrt{2}} = \frac{\sin x}{2 \sin x + \sqrt{2}}$$

$$\frac{\cos 3x - \sin x}{2 \sin x + \sqrt{2}} = 0$$

$$\sin x = \cos\left(\frac{\pi}{2} - x\right)$$

$$\frac{\cos 3x - \cos\left(\frac{\pi}{2} - x\right)}{2\sin x + \sqrt{2}} = 0$$

$$\left[\begin{aligned} \cos \alpha - \cos \beta &= 2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\beta - \alpha}{2}\right) \\ \cos \alpha + \cos \beta &= 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \\ \sin \alpha + \sin \beta &= 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \\ \sin \alpha - \sin \beta &= 2 \sin\left(\frac{\alpha - \beta}{2}\right) \cos\left(\frac{\alpha + \beta}{2}\right) \end{aligned} \right.$$

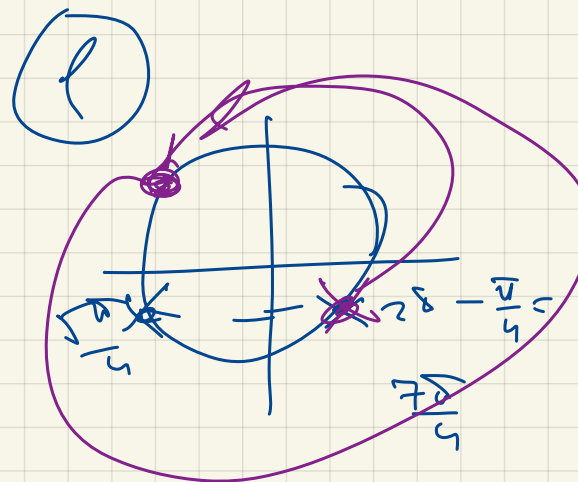
$$\frac{2 \sin\left(\frac{3x + \frac{\pi}{2} - x}{2}\right) \sin\left(\frac{\frac{\pi}{2} - x - 3x}{2}\right)}{2\sin x + \sqrt{2}} = 0$$

$$\sin\left(\frac{2x + \frac{\pi}{2}}{2}\right) = 0$$

$$\sin\left(\frac{\frac{\pi}{2} - 4x}{2}\right) = 0$$

$$\sin x \neq -\frac{\sqrt{2}}{2} \leftarrow$$

$$\frac{2x + \frac{\pi}{2}}{2} = \pi k$$



①

$$2x + \frac{\pi}{2} = 2\pi k$$

$$2x = -\frac{\pi}{2} + 2\pi k$$

$$x = -\frac{\pi}{4} + \pi k \Rightarrow \left(\frac{3\pi}{4} + 2\pi k \right)$$

②

$$\frac{\frac{\pi}{2} - 4x}{2} = \pi k$$

$$\frac{\pi}{2} - 4x = 2\pi k$$

$$4x = \frac{\pi}{2} - 2\pi k$$

$$x = \frac{\pi}{8} - \frac{2\pi k}{2}$$

⑥

$$\frac{S}{8} \cdot 8 + 0,18 S \cdot 4 - \frac{S}{8} \cdot 0,18 (1+2+3)$$

$$- 0,14 \frac{S}{8} (4+5+6+7) + 1,14 \frac{S}{8} =$$

$$+ 4 \cdot 0,14 S$$

$$S + 0,72 S + 0,14 25 S - 0,135 S$$

$$- 0,385 S + 0,56 S = 2200$$

$$1,9025 S = 2200$$

$$\left(\sqrt{12-x^2} - y \right) \left(\frac{(x+y)^2 + (y+y)^2 - 8(x+y) + x^2 - y^2 - 2y}{x^2 - y^2 - 2y} \right)$$

$$-6x^2 - 48x - 24$$

$$x^2 + 8x + 16 + x^2 \Rightarrow \underline{2x^2 - 24 + 8y}$$

$$\underline{16 + 16 - 32 - 24}$$

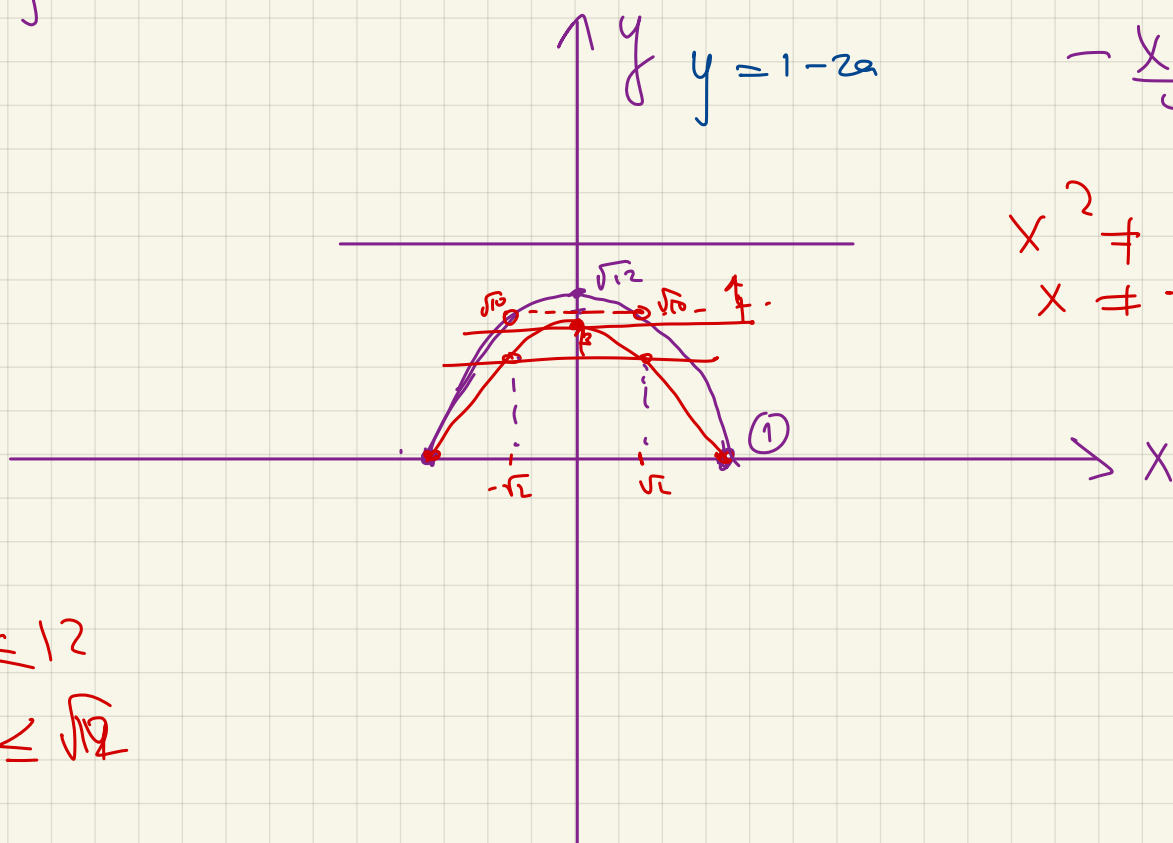
$$\left\{ \begin{array}{l} \frac{(\sqrt{12-x^2} - y)(2x^2 - 24 + 8y)}{2-x^2} = 0 \\ y = 1-2a \end{array} \right.$$

$$\frac{-2x^2 + 24}{8} = y$$

$$-\frac{x^2}{4} + 3 = y$$

$$x^2 + 2$$

$$x \neq \pm \sqrt{2}$$



$$x^2 \leq 12$$

$$-\sqrt{12} \leq x \leq \sqrt{12}$$

$$y = \sqrt{12 - x^2}$$

$$y \geq 0$$

$$a \in \left\{ \frac{1}{2} \right\} \cup \left\{ \frac{5}{2} \right\}$$

$$\cup (5; \sqrt{10}) \cup (\sqrt{10}; \sqrt{2})$$

$$y^2 = 12 - x^2$$

$$y^2 + x^2 = 12 \quad \sqrt{12} = R$$

$$y = 0 \quad a = \frac{1}{2}$$

$$y = -\frac{2}{9} + 3 = 3 - \frac{1}{2} = \frac{5}{2}$$

$$\frac{5}{2} = 1 - 2a \quad a = -\frac{3}{4}$$

$$y = \sqrt{12 - 2} = \sqrt{10}$$

$$1 - 2a = \sqrt{10} \Rightarrow a = \frac{1 - \sqrt{10}}{2}$$

$$\left(y - \sqrt{10 - x^2} \right) \left((x+5)^2 + (y+5)^2 - 10 \mid x+2, 5 \right) + x^2 - y^2 + 5 = 0$$

$$\sqrt{x^2 - 1}$$

$$x = -1$$

$$y = a + a - 1 \Rightarrow \underline{a(x+5) - 1} \quad (-1; -1)$$

$$\underline{x^2 - 1 > 0}$$

$$\underline{-1 > x > 1}$$

$$2x^2 + 10y - 20$$

$$y = \frac{-2x^2 + 20}{10} = -\frac{x^2}{5} + 2$$

$$f(-1) = -\frac{1}{5} + 2 = +\frac{9}{5}$$

$$x^2 = 10$$

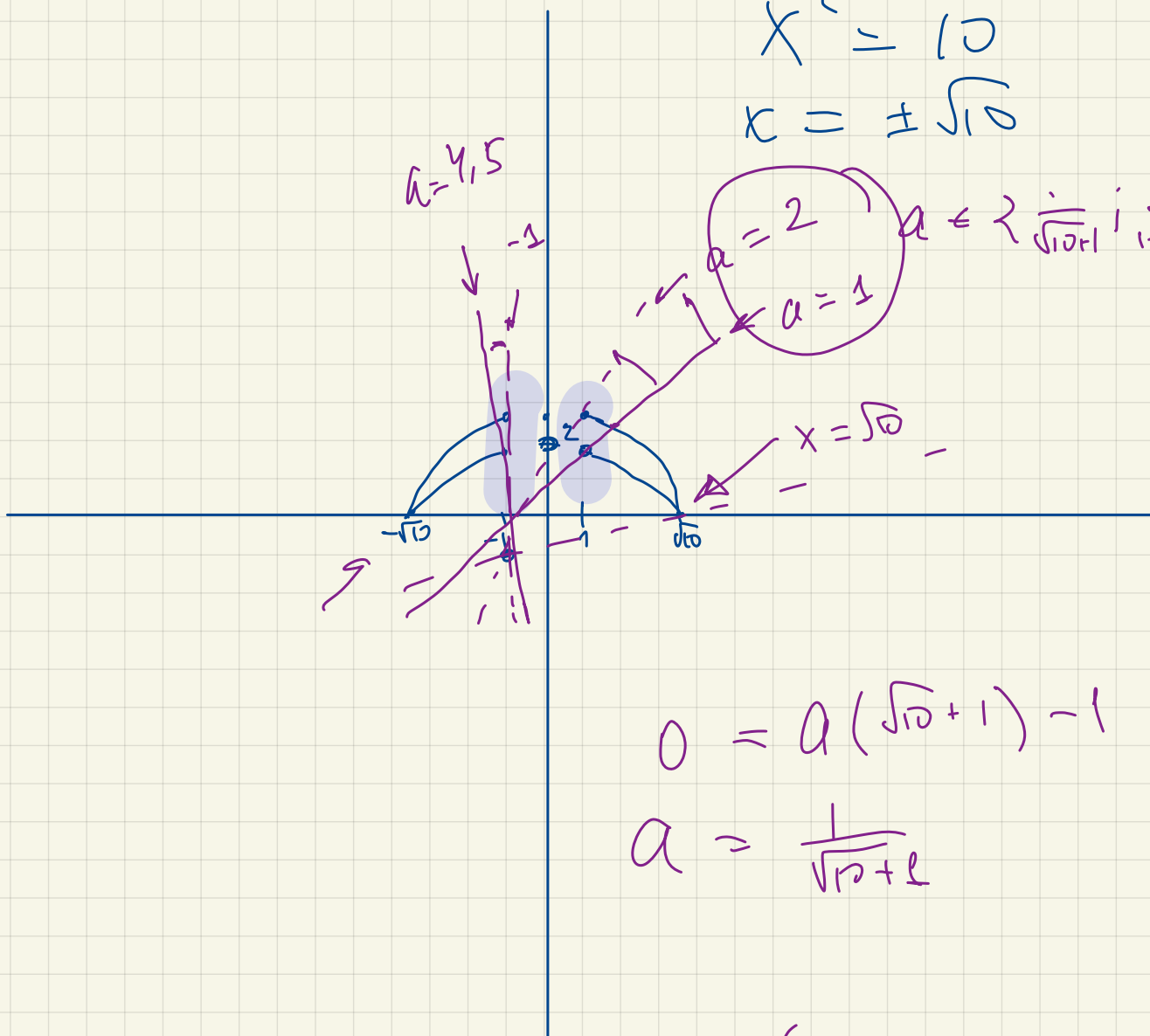
$$x = \pm\sqrt{10}$$

$$h = 4,5$$

$$a = 2$$

$$a = 1$$

$$a \in \left\{ \frac{1}{\sqrt{10}+1}, \frac{1}{1-\sqrt{10}} \right\}$$



$$0 = a(\sqrt{10}+1) - 1$$

$$a = \frac{1}{\sqrt{10}+1}$$

$$y(-2) = \sqrt{10-1} = 3$$

$$(-1; 3)$$

$$3 = a(0) - 1$$

$$3 \neq -1$$

$$(-1; \frac{9}{5})$$

$$\frac{a}{s} = a(0) - \frac{1}{2}$$

$$\frac{a}{s} \neq -1$$

